

Summary

Score

Reviewer	Role	Expertise	Rating	
R1	Reviewer	Knowledgeable	Rather Accept	4
R2	Committee Member	Expert	Rather Accept	4
R3	Reviewer	Passing Knowledge	Definitely Reject	1

Reviewer 2 (Committee Member, Expert) — Rather Accept

Strengths

- **The strongest contribution is the cross-context empirical comparison** of anchoring strategies. The paper usefully demonstrates that mobile AR design should not always maximize content visibility — accessibility, interference, and predictability must be balanced by task context. *(This framing is valuable for rebutting R3's "common sense" critique.)*
- Well-structured multi-stage evaluation: pilot identifies limitations of conventional anchors → four new techniques designed in response → two complementary studies (individual mobile multitasking + collaborative sensemaking) provide broader empirical basis than a single-task comparison.

Weaknesses — Must Fix

- **AdaptiveLock description contradicts itself** between §5.1.2 and Study 1 discussion. *(Also raised by R1 — undeniable error.)*
- **Novelty positioning needs reframing**: the individual techniques build on existing anchoring, peripheral display, and motion-aware AR work. **The contribution should be positioned as an empirical/design contribution (the cross-context comparison and resulting design implications), not as fundamentally new anchoring mechanisms.** The work also needs to more explicitly distinguish itself from prior walking AR anchoring and motion-aware AR interface studies.
- **Acknowledgment section is present in the anonymous version** — should be removed for the review version (or AI disclosure moved to a dedicated disclosure/ethics statement) and restored for camera-ready.

Weaknesses — Address with Partial Concession

- **Environment order is not counterbalanced in Study 1:** the four techniques used a Latin Square, but the three environments (Open Space → Narrow Corridor → Room-to-Room) were always presented in fixed order. Combined with 3–4.5 hour sessions, this confounds environment effects with order, learning, fatigue, adaptation, and prolonged HMD use. *Strategy: acknowledge as limitation; downgrade environment-related findings to exploratory; explain the escalating-difficulty rationale (safety, complexity); commit to substantially weakening causal claims about spatial context.*
- **No traditional baseline (world/head/body-lock) in main studies** — only in the pilot. Including conventional baselines in the main experiments would clarify how much the proposed techniques improve over standard anchoring. *Strategy: clarify that main studies evaluate relative differences among the proposed techniques; commit to softening "outperformed" language in Abstract and Introduction; note that the pilot established baseline comparisons. Offset*
- **Small N=12 in Study 1** for a 4×3 within-subjects design limits robustness and generalizability of statistical results. *Strategy: note that N=12 within-subjects is standard for HCI/AR studies (Study 2 has N=24/12 dyads); commit to adding confidence intervals and/or robustness checks; temper generalization claims.*

Bottom Line for R2

R2 is the most professional and balanced reviewer, and as committee member carries the most weight. None of R2's concerns require new experiments — all can be addressed through textual revisions and softening of claims. R2's framing of "cross-context empirical comparison" as the strongest contribution is the most valuable line of defense in the entire review set.

Reviewer 1 (Knowledgeable) — Rather Accept

Strengths

- Topic is timely, important, and well-motivated for the mobile AR field
- Study descriptions are clear and well-supported by appropriate figures
- Multiple studies provide insights across differing mobility contexts and cognitive demands
- Data collection is appropriate; analysis appears conscientious and accurate
- Authors are appreciated for releasing source code to support reproducibility

Weaknesses — Must Fix

- **AdaptiveLock description is internally inconsistent:** §5.1.2 says "body-locked walking / world-locked stationary," but §6.4 reverses this. *(Also raised by R2 — undeniable error.)*
- **Related Work §2.3 (Anchoring Techniques) is too thin** for what is the paper's central topic. R1 names five papers that should be incorporated: Lu 2023 (Glanceable AR, SUI), Lindlbauer 2019 (Context-Aware MR, UIST), Pfeuffer 2021 (ARtention, C&G), Jannat 2025 (Body-Anchored AR, ISMAR), Satkowski 2022 (Above & Below, ISMAR). Path-anchoring (ref [27]) should also be discussed in §2.3 rather than only cited elsewhere.
- **General Discussion (§8) completely lacks engagement with prior work** — only the authors' own findings are discussed. Specifically calls out the need to relate findings to Lindlbauer's context-aware adaptation, Lu's glanceable AR, and Jannat's body-anchoring work. This is one of the two named "major issues."
- **Design Guidelines §8.4 are redundant and overlapping:** Environmental Complexity Assessment + Cognitive Load Detection $\approx$ User Attention State Monitoring; Task Phase Recognition is a temporal aspect of Cognitive Load; Mobility Pattern Analysis overlaps with the others. The five items need to be reorganized into genuinely orthogonal dimensions.
- **AdaptiveFlow trigger conditions are unclear:** does fast walking alone trigger the world-locked transition, or only head yaw above threshold? The text and video give conflicting impressions. Needs rephrasing plus design rationale.
- **AdaptiveLock vs. existing locomotion-aware anchoring** (path-anchoring, ref [27]) — relationship needs to be explicitly articulated, or renamed if the same.
- **HoloLens 2's limited FoV** is not acknowledged in Limitations but very likely influenced spatial awareness. *(Also implied by R3.)*
- **Overgeneralized statements without backup** in Introduction and Pilot Study sections — e.g., "Body-based anchors keep content in the central visual field" is not universally true (they can also be peripheral). Should be rephrased to "our body-based anchor..." or backed with citations.
- **Chang et al. [4] on-demand hiding mechanisms** are not discussed in Limitations.
- **PeriphGuide description final 1–2 sentences** need rewriting; the last sentence does not make sense as written.

Weaknesses — Address with Partial Concession

- **Writing has GenAI-style overgeneralization** ("sentences that sound good but say nothing when examined"). The Acknowledgment already discloses GenAI use for writing refinement, so this concern is grounded. *Strategy: commit to a careful manual editing pass; do not concede that AI generated substantive content.*
- **§7.4 begins interpretation and §7.5 begins cross-study comparison** before the General Discussion. Should be streamlined. *Strategy: minor structural revision; this is a common HCI convention but can be tightened.*
- **Fig 2b semi-transparent panel** raises a question about how gaze "through" the panel was counted for Interface Attention Ratio. *Strategy: clarify that AOI classification is*

based on gaze-ray intersection with 2D panel bounds; transparency does not affect AOI assignment.

Bottom Line for R1

R1 explicitly states the paper can be accepted if the Discussion and Related Work are thoroughly improved before the camera-ready deadline. The path to satisfying R1 is clear and entirely additive — no new experiments required.

Reviewer 3 (Passing Knowledge) — Definitely Reject

R3 self-identifies as having only "Passing Knowledge" of the area, and several criticisms rest on misreading specific implementation details or on dismissing a key finding (the preference–performance dissociation) as an implementation flaw. The rebuttal should be respectful but firm with direct textual evidence.

Strengths (acknowledged by R3)

- The paper's conclusions are "quite likely correct"
- Pilot study is robust for a pilot
- The dynamic anchoring suggestion is "probably true"

Weaknesses — Address with Partial Concession

- **Long 3–4.5 hour sessions introduce fatigue** that may impact memory recall even with counterbalancing. (*Also raised by R2.*) *Strategy: point to existing mitigations in §6.2.3 — 5-min breaks per condition, 30-min breaks per set, recalibration available; acknowledge as remaining limitation.*
- **Temporal Stability/Availability construct is only defined in the Appendix**, hurting readability. *Strategy: add brief definition in main text §6.2.4 — easy fix.*
- **HoloLens 2 FoV limitations** implicitly criticized via the broader implementation concerns. (*Also raised by R1.*) *Strategy: add to §9 Limitations.*

Weaknesses — Must Rebut

- **"Head-locked could have used smoothing filter / thresholding — not explored."**
REBUT: §5.1.1 explicitly implements temporal smoothing for both PeriphGuide ("per-frame smoothing, Lerp = 0.1") and ScaleStable ("temporal smoothing, Move/Rotate Lerp = 0.1"). R3 missed this detail in the implementation section.

- **"Head-locked FoV is massive — setting up PeriphGuide for failure for navigation."** **REBUT:** PeriphGuide's FoV occupancy ($\pm 40^\circ\text{H}$ / $\pm 30^\circ\text{V}$ within the 52° diagonal FoV) is integral to its design definition — peripheral clamping at FoV boundaries. Equalizing FoV across techniques would eliminate the very design contrasts the study is evaluating.
- **"AdaptiveLock uses smaller FoV — unfair comparison."** **REBUT:** Same principle as above. FoV parameters (AdaptiveLock walking: $\pm 30^\circ\text{H}$, $\pm 20^\circ\text{V}$; panel distance 0.85m vs. ScaleStable's 0.5–3.5m range) are constitutive of each technique's design rationale, not confounds. The study evaluates trade-offs *between* these design choices, not a controlled-FoV ablation.
- **"PeriphGuide is badly named — not actually peripheral, almost entirely fills FoV."** **PARTIAL REBUT:** $\pm 40^\circ\text{H}$ / $\pm 30^\circ\text{V}$ positions content at the FoV edges by definition, within HoloLens 2's 52° diagonal (43°H , 29°V) FoV. Can clarify naming rationale in §5.1.1; not a substantive issue.
- **"The anchoring trade-off section just seems like common sense."** **REBUT using R2:** The committee member explicitly identifies cross-context empirical comparison as "the paper's strongest contribution." A committee member with Expert self-rated expertise disagrees with R3's "common sense" characterization.
- **"SUS is just a bad questionnaire."** **REBUT:** SUS is a standard, widely-validated HCI usability instrument used across decades of literature. We triangulated with custom 7-point Likert scales, preference rankings, and qualitative interviews — not relying on SUS alone.
- **"User preference for ScaleStable implies other methods were poorly implemented."** (*R3 repeats this point twice — it is the core of the rejection.*) **REBUT:** This is a misinterpretation of a deliberate finding. The preference–performance dissociation (users prefer ScaleStable despite its measurably lower recall and interface attention) is itself a contribution, articulated in §6.4. Users prioritize minimizing visual interference even at a measurable performance cost — this is the finding, not an artifact.
- **"How is mobile AR fundamentally different from VR navigation?"** **REBUT briefly:** Mobile AR requires simultaneous awareness of physical environment for safety and task-relevant perception — a constraint absent in VR where the virtual world is the only relevant environment. Can add 1–2 sentences in Introduction for clarity.
- **"RQ2 is not really a research question — too vague, no hypothesis."** **PARTIAL REBUT:** RQ2 is intentionally exploratory (how the accessibility–stability trade-off manifests under different demands), not hypothesis-driven. This is standard for context-comparison studies. Can sharpen wording. **Our RQ2: How does the accessibility-stability trade-off play out when AR interfaces support complex environmental processing and collaborative cognition?**
- **"No significant differences between the other conditions besides SS — safe to assume?"** **CLARIFY:** §6.3 post-hoc tests already report this. The pattern itself is informative: the three accessibility-supporting techniques cluster together while ScaleStable is distinct — this is the finding, not a study weakness.
- **"Recommend re-implementing UIs and running a smaller supplementary study."** **REBUT:** This recommendation depends entirely on the implementation critiques above

(smoothing, FoV), which rest on misreadings of §5.1.1. The design tensions are intentional features of the study, not implementation flaws.

Bottom Line for R3

R3's rejection is grounded primarily in misreading §5.1.1 (smoothing implementation) and in re-interpreting a deliberate dissociation finding as an implementation defect. R3 has Passing Knowledge expertise and is contradicted on the central "common sense" point by the Expert committee member (R2). The rebuttal should concede genuinely useful points (Temporal Stability definition, fatigue acknowledgment, FoV limitation) while firmly correcting the misreadings using direct textual evidence from §5.1.1, §6.3, and §6.4.

Cross-Reviewer Convergence (the most credible criticisms)

These are the issues raised by more than one reviewer — the rebuttal should open by addressing these because they carry the most weight:

- **AdaptiveLock §6.4 description error** — R1 + R2 (undeniable; correct immediately)
- **Novelty needs reframing as empirical/design contribution** — R1 (implicit) + R2 (explicit)
- **Long sessions / fatigue concern** — R2 + R3 (acknowledge; point to existing mitigations)
- **Environment / generalization concerns** — R2 (order confound) + R3 (fatigue impact) — same underlying worry from different angles
- **HoloLens 2 FoV limitation missing** — R1 (explicit) + R3 (implicit via implementation critique)

Questions

1. The reason for ISMAR submission is for IEEE TVCG journal paper, the score makes it seems impossible
2. Some of Reviewer 3's comments are due to misreading, how should I address this?
3. Reviewer 1 and 3 both mentioned FoV limitation, however, this is one of the main contribution, as for the baseline (head-, body-, world-locked) also have inherently different FoVs by design. Importantly, we deliberately controlled for visual size identical across all four designs, both during on-the-go and when stationary. How should we respond to this in the rebuttal?

Original Comment

Reviewer 2 (committee member)

Expertise

Expert

Contribution

This paper investigates how AR information panels should be anchored during mobile use, where users must balance access to virtual content with attention to the physical environment. Based on a pilot study comparing conventional world-, head-, and body-locked panels, the authors propose four anchoring techniques: ScaleStable, PeriphGuide, AdaptiveLock, and AdaptiveFlow. These techniques explore different ways to balance spatial stability, temporal accessibility, and visual interference. The paper evaluates the techniques in two studies: an individual walking-and-memory task and a remote collaborative sensemaking task. The main contribution is showing that anchoring preferences depend on task context: accessible/adaptive techniques help during sustained mobility and information recall, while stable and less visually intrusive anchoring is preferred during collaborative reasoning and environmental inspection. The novelty is moderate: the individual anchoring ideas build on prior work, but the cross-context comparison and the resulting design implications for mobile AR anchoring are useful.

Paper Length

The paper length matches its contribution

Significance

Good

Originality

Good

Research Quality

Good

Technical Correctness

Acceptable

Writing Style & Organization

Good

Overall Rating

Rather accept: But I would not argue for accepting this submission.

Detailed Rationale

Strengths:

1. Meaningful design insight. The paper's strongest contribution is its cross-context empirical comparison of anchoring strategies. It usefully shows that mobile AR design should not always maximize content visibility; instead, accessibility, interference, and predictability should be balanced depending on task context.

2. Well-structured multi-stage evaluation.

The evaluation is well structured: the authors first use a pilot study to identify limitations of conventional world-, head-, and body-locked anchors, then design four techniques based on these observations, and finally evaluate them in two complementary studies. Study 1 examines individual mobile multitasking, while Study 2 extends the evaluation to collaborative sensemaking, giving the paper a broader empirical basis than a single-task comparison.

Weaknesses:

1. Originality and relationship to prior work.

a) The contribution is useful but somewhat incremental, since the individual techniques build on existing anchoring, peripheral display, and motion-aware AR ideas. The authors should more clearly position the novelty as an empirical/design contribution rather than as fundamentally new anchoring mechanisms. They should also more explicitly distinguish the work from prior walking AR anchoring and motion-aware AR interface studies.

2. Methodological concerns.

a) Fixed environment order: In Study 1, the four techniques were counterbalanced with a Latin Square, but the environments were always presented in the same order: Open Space, Narrow Corridor, and Room-to-Room. While this is understandable for safety and increasing task complexity, it confounds environment effects with order effects such as learning, fatigue, adaptation, and prolonged HMD use. Given the long 3–4.5 hour sessions, the environment-related findings should be interpreted more cautiously. The authors should validate these findings with a counterbalanced environment order. If this is not feasible, they should explicitly acknowledge this as a limitation, treat the environment-related results as exploratory, and substantially weaken causal claims about the effects of spatial context.

b) Small sample size in Study 1: Study 1 includes only 12 participants in a 4×3 repeated-measures design, meaning each participant completed 12 conditions. This provides useful within-subject comparisons, but the small N may limit the robustness and generalizability of the statistical results. The authors should be cautious in generalizing these findings and consider reporting additional uncertainty information, such as confidence intervals or robustness checks.

3. Evaluation.

a) Lack of formal baseline comparison: The pilot study compares conventional world-, head-, and body-locked anchors, but the two main studies only compare the four proposed techniques. Including conventional baselines in the main experiments would make it easier to assess how much the proposed techniques improve over standard anchoring approaches. At minimum, the authors should clarify that the main studies evaluate relative differences among the proposed techniques rather than definitive improvements over traditional anchors.

4. Writing issues.

a) Some technique descriptions are inconsistent: The paper should carefully check the description of AdaptiveLock. Earlier, AdaptiveLock is described as switching to world-locked when stationary and body-locked while walking. However, in the Study 1 discussion, the description appears reversed, saying it is “body-locked when stationary” and “world-locked during movement.” This inconsistency can confuse readers and should be corrected.

b) Minor double-blind / formatting issue: The anonymous version includes an Acknowledgment section. It does not appear to reveal author identity, since it mainly thanks volunteers and discloses the use of generative AI tools and IKEA Room Design. Still, acknowledgments are often omitted during double-blind review and restored in the camera-ready version. The AI-tool disclosure could be moved to a dedicated disclosure or ethics statement.

Reviewer 1 (reviewer)

Expertise

Knowledgeable

Contribution

The paper presents and assesses performance of four different anchoring techniques for always-on information displays in AR on-the-go scenarios, where locomotion and stationary phases might constantly switch. While none of the utilized reference frames are novel, most related works only investigate the placement at one of the reference frames for on-the-go use cases so far, creating a niche the authors correctly identified. By adapting the reference frame to the current walking situation, the authors address a timely issue and expand on recent works in the field. Furthermore, the authors provide design guidelines for on-the-move always-on interfaces based on the current mobility and cognitive demands of the user.

Paper Length

The paper is too long to address its claimed contribution

Significance

Good

Originality

Acceptable

Research Quality

Acceptable

Technical Correctness

Very Poor

Writing Style & Organization

Poor

Overall Rating

Rather accept: But I would not argue for accepting this submission.

Detailed Rationale

I want to start by thanking the authors for their work on this timely and important topic.

For quick reference, I summarized my assessment of the paper into strong points, weak points and major issues. The detailed review organized by section can be found below.

Strong Points:

- The paper's motivation is straightforward and well organised.
- Overall, the study descriptions are clear and well supported by appropriate figures.
- Multiple studies are introduced to gain insights based on differing mobility contexts and cognitive demands.
- Data collection is appropriate and analysis seems conscientious and accurate

Weak Points:

- The overall writing style often gets in the way of understanding. This might be due to use of generative AI
- Certain design choices are unclear or underspecified
- Often statements seem too broad / overgeneralized or are not sufficiently backed up by RW

Major Issues:

- RW is very shallow, especially on the central topic of anchoring techniques

- Discussion completely lacks any engagement with RW

Overall:

I often stumbled across sentences which were either way to overgeneralized or sounded good during the first read but didn't really say anything when thinking about them. These usually are issues connected to usage of generative AI. I highlighted some occurrences but want to urge the authors to carefully go through the paper again and improve any occurrences.

Introduction:

Overall, the topic is clear and well motivated. However, in the second paragraph two broad statements are made and not properly backed up in my opinion:

- "While these techniques are simple to understand and work well in many scenarios, they prove less sufficient for the complex demands under mobile AR usage [21]." – The statement is too broad to back it up with just one source which investigates one mobile AR usage. I suggest finding further RW or rephrasing.

- "Traditional anchoring approaches [...] can become insufficient to adapt to these fluctuating priorities" – Why? This is not clearly explained nor backed up by any source. I suggest careful rephrasing.

The third to fifth paragraphs could be shortened to avoid repetition within the paper. For example, it is difficult to evaluate how one anchoring style is better than another without a deeper understanding of the anchoring styles. Similarly, it is difficult to interpret the individual findings of Study 1 or Study 2 without knowing the specifics of the study.

This could also be a chance to provide a clear overview of key contributions of the paper which are right now hidden in the last three paragraphs.

Related Work:

The related work revolves around Remote Collaborative Systems in AR, Multitasking and Attention Management as well as Anchoring Techniques. To my surprise, anchoring techniques were only a short part of the RW, while being the main topic. Furthermore, this part stops at a shallow description of world, body and head anchoring, before circling back to multi-user collaboration. Locomotion adaptations like path-anchoring[ref 27] are omitted, as well as any presentation of current context aware [b] or on demand techniques[a,d]. Please consider the following list of related works which should be included and presented here in section 2.3. This not only provides a clearer overview for the reader but also allows the authors to position their work better within these related works. Furthermore, the results of this work should be discussed in relation to these works, with [c,e] potentially allowing interesting comparisons.

a. Feiyu Lu, Leonardo Pavanatto, and Doug A. Bowman. 2023. In-the-Wild Experiences with an Interactive Glanceable AR System for Everyday Use. In Proceedings of the 2023 ACM Symposium on Spatial User Interaction (SUI '23). Association for Computing Machinery, New York, NY, USA, Article 11, 1–9. <https://doi.org/10.1145/3607822.3614515>

(an earlier work is already cited, this one might be more appropriate for some discussions)

b. David Lindlbauer, Anna Maria Feit, and Otmar Hilliges. 2019. Context-Aware Online Adaptation of Mixed Reality Interfaces. In Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology (UIST '19). Association for Computing Machinery, New York, NY, USA, 147–160.
<https://doi.org/10.1145/3332165.3347945>

c. Ken Pfeuffer, Yasmeen Abdrabou, Augusto Esteves, Radiah Rivu, Yomna Abdelrahman, Stefanie Meitner, Amr Saadi, Florian Alt. 2021. ARtention: A design space for gaze-adaptive user interfaces in augmented reality. In Computers & Graphics, Volume 95, Pages 1-12, ISSN 0097-8493.
<https://doi.org/10.1016/j.cag.2021.01.001>

d. M. -E. Jannat, S. M. Shah and K. Hasan, "Exploring Body-Anchored Augmented Reality Interfaces Across Different Mobility and Social Contexts," 2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Daejeon, Korea, Republic of, 2025, pp. 1429-1439, doi: 10.1109/ISMAR67309.2025.00147.

e. M. Satkowski, R. Rzayev, E. Goebel and R. Dachsel, "ABOVE & BELOW: Investigating Ceiling and Floor for Augmented Reality Content Placement," 2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Singapore, Singapore, 2022, pp. 518-527, doi: 10.1109/ISMAR55827.2022.00068.

Pilot Study:

The pilot study is kept brief with details being provided in the appendix, which I appreciate. Nonetheless, a brief description of the task – beyond walking—reading—communicating - would improve the reading flow and the understanding of the pilot.

When describing the found factors, there again are overly generalized statements being made:

- „Body-based anchors keep content in the central visual field” – This is not necessarily true, as body-based anchors also can provide peripheral interfaces, e.g. displayed on the floor or in the sky. I would suggest rephrasing, e.g. our body-based anchor

- “During movement, environmental processing demands increase while cognitive bandwidth for locating distant content decreases—world-anchored content becomes

not only less accessible but actively disruptive“ – Either needs a source for backup or a careful rephrasing, e.g. participants stated that... .

Adaptive Anchoring Design:

The last 1-2 sentences in the description of PeriphGuide need an improvement, especially the last one does not make sense.

Based on the description, AdaptiveLock sounds very similar to path-anchoring and even seems to behave similarly in the supplementary video. If it is the same it needs to be named accordingly to avoid confusion for the reader, if not key differences should be pointed out here.

The description of AdaptiveFlow is not completely clear to me. From the teaser image and the supplementary video, I assumed that it only anchors to the world briefly on quick head movements. Here it sounds like fast walking triggers world anchoring, too? Should be rephrased for clarity. And if fast walking also triggers this, what is the design choice behind this?

At this point I'd like to thank the authors for their efforts to provide the source code to the community, which not only eases reproducibility but also fosters prototyping efforts.

Study 1:

In fig. 2b the interface is slightly transparent, allowing participants to observe the environment through it. I wonder how gazing "through" the Slide Deck for navigational cues was counted regarding the interface attention ratio. Was this issue considered and taken into account for the gaze time?

Furthermore, the description of AdaptiveLock provided in 6.4 is the exact opposite of 5.1.2 and needs fixing.

Study 2:

In the 7.4 results section the authors already start to interpret the data, the 7.5 study 2 discussion starts to make comparisons to study 1 before the general discussion. This should be streamlined.

General Discussion:

One of the major issues of this paper is the complete lack of engagement with related works in the discussion. Context aware interfaces by Lindlbauer et. al.[b], glanceableAR by Lu et al.[a] or different body-anchoring as described by Jannat et al.[d] are just some naturally emerging topics to discuss.

The authors need to rework the discussion sections to put their findings into context of the research field. Only discussing their own results without putting it into relation to other works is completely insufficient, especially as they touch on the topics of the other works in their own discussion.

The design guidelines in 8.4. seem known already and are not well described or redundant, e.g. Environmental Complexity Assessment and Cognitive Load Detection taken together are the same as User Attention State Monitoring, Task Phase Recognition results from Cognitive Load Detection and Mobility Pattern Analysis. As this has a significant impact on the claimed contributions, I would strongly advise engaging with the related work again to greatly improve the discussion and design guidelines.

Limitations:

The Limitations section also falls short on a few points:

- The cited Chang paper[4] suggests that on-the-go interfaces provide mechanisms to hide them on demand, which is not discussed nor put as a limitation.

- Glanceable techniques and/or other on demand techniques could have improved cognitive load or interface interference during stationary scenarios.

- The very limited FoV of the Hololens2 most likely influenced spatial awareness

Recommendation:

In conclusion, the topic is well motivated, timely and important. The studies and analysis appear sound and appropriate. However, the paper demonstrates a general lack of engagement with related work, particularly in the discussions, and is often overly generalised. This has a significant impact on the paper's claimed contribution, as comparisons with or positioning within related work are hardly possible. However, I believe these issues can be resolved within the time until the camera ready deadline.

I would rather accept this paper on the terms that the discussion and related work are thoroughly improved upon for the camera ready version.

Furthermore, I would like to encourage the authors to continue their work on this important and timely topic. There is definitely a need for more research into how interfaces and interactions can and should work for AR on-the-move!

Reviewer 3 (reviewer)

Expertise

Passing Knowledge

Contribution

This paper runs a series of studies to assess different AR interfaces for managing content placement relative to users. They introduce four methods for this, and simulated AR to study their methods in different tasks.

Primarily, they assess: World-Locked vs Head-Locked vs Body-Locked methods.

Paper Length

The paper length matches its contribution

Significance

Poor

Originality

Acceptable

Research Quality

Poor

Technical Correctness

Poor

Writing Style & Organization

Poor

Overall Rating

Definitely reject: I would argue strongly for rejecting this submission.

Detailed Rationale

Framing of the problem: In mobile AR, people have to process physical and digital information while navigating the real-world environment. I'm reading an implication that this works in "non-AR" but not in AR; How is this fundamentally different from VR interfaces though, where users have to navigate the virtual world?

There are two studies that have conflicting findings, which they say suggests that AR systems should consider dynamic anchoring strategies (probably true).

Pilot Study: The pilot study is pretty robust for a pilot study, although from the findings in Table 1 (Appendix) it's very difficult to get a clear idea of how users performed on the task. I fear that results are entirely dependent upon implementation details; for example, the head-locked is rated as "low spatial predictability" because, essentially, it jitters quite a bit, but a smoothing filter for the motion could have very easily been

used, or even thresholding before moving the UI. It just seems like this was not explored.

The anchoring trade off section just seems like common sense.

Another issue is that the head-locked FOV is massive, so of course it's going to cause occlusions that make navigation difficult. It's sort of setting up PeriphGuide for failure for navigation.

AdaptiveLock is using a smaller FOV and thus seems difficult to compare properly.

PeriphGuide is badly named, since it's not really a peripheral UI, but one that takes up almost the entirety of the FOV.

Participants were tasked with remembering items on the UI Canvas, to isolate how the anchoring techniques affected their recall.

3-4.5 hours per participants sounds like it'll introduce quite a lot of fatigue. I don't understand why each set / trial seems to have taken so long that the sessions needed to take this long?

But such a long study time seems like it'll certainly impact the memory recall - even with counterbalancing, it's hard to know how impacted they can be by individual participant fatigue.

Study 1 Results:

ScaleStable, by being locked to one point, will certainly have lower recall. It's safe to assume there were no significant differences between the other conditions?

Temporal Stability being defined only in the appendix hurts the readability of this, but I'm surprised that the only significant difference is between PG and SS.

The rest of the results seem to make sense to explain ScaleStable's results, but the study provides few insights about the other methods, although the preference seemed to be for ScaleStable and AdaptiveLock.

The most likely explanation for the lack of differences in SUS despite poor performance is that SUS is just a bad questionnaire.

The user preference scale seems to imply not that they liked SS most, but that the others were just poorly implemented by having too-large FOVs.

I don't doubt that AdaptiveLock is probably better than SS or even PeriphGuide, but doubt the implementation allows a thorough exploration of the comparison. This is even sort of acknowledged in the discussion.

Is RQ2 really a research question? This is intentionally vague and it's unclear what, if any hypothesis, exists.

Once again, SS seems to achieve the highest user preference, once again indicating to me that it was poor design of the other methods that made them so obtrusive to users and their remote partner.

Overall:

I think the paper's conclusions are quite likely correct, but I don't think this was rigorously evaluated, starting at the implementation of the methods. This makes it

very difficult to prove the point the authors try to make. I recommend the authors reevaluate how they've developed the UIs (make UI consistent, have some headset motion smoothing to prevent jitter) and more scientifically evaluate them with a smaller study to supplement the current ones.

ISMAR Meta review

Dear authors,

We have held an extensive discussion regarding the submission. The overall conclusion of this discussion is that the paper cannot be accepted in its current form. The reviewers recognized several strengths: the topic is timely and relevant (2AC, R1, R3), the study offers meaningful design insights (2AC, R1), and the mobile environments provide high ecological validity (2AC, R1, R3). However, the following key concerns prevented an acceptance:

- Methodological and implementation issues: There are concerns regarding the rationale behind the proposed placement strategies (e.g., ScaleStable), the impact of environment presentation order on results, and potential flaws in the implementation of UIs of fixed angular size, and gaze measurements (2AC, R1, R3).
- Lack of engagement with related work: Reviewers note that the Related Work does not sufficiently cover existing literature and the Discussion also lacks engagement with prior work (R1, R3). 2AC goes even further and notes that, given the current positioning of the work, proposed placement strategies feel somewhat incremental (2AC).
- Insufficient rebuttal: the provided response was deemed too brief and failed to provide a concrete plan for how the manuscript would be restructured or how the proposed techniques relate to existing work (R1, R3).

I encourage the authors to carefully reread all the comments, improve the manuscript by addressing these issues, and resubmit it elsewhere.